

Exercise

1. Consider a database, which contains the details of human resource for a company with four relations Location, Job details, Department and Employee given in pages 4, 5, and 6. The relations show some values for human resource database of a company. The relations are uniquely identified by the attributes Location Id, Job Id, Department Id and Employee Id respectively. Each of the departments are located in different places and employees are serving in various job roles in different departments.
 - (a) Create a **user** with your *index number* as the user name at '*localhost*' and give **all privileges** to the created user.
 - (b) List down the details of grants for the created user.
 - (c) Login using the created user to answer the questions ahead.
 - (d) Create a database **Human_Resource** with the above mentioned four relations by clearly specifying the corresponding **primary key** and **foreign key** constraints in addition to the conditions described in **Table 1** for each of the relations. Populate the relation with the data given in the shared folder or in the Tables.
 - (e) Write MySQL statements to **create view** for each of the following cases:
 - i. To display the *First_name*, *Last_name*, *Department_Id*, and *Department_name* for all employees who are working in the department with *Department_Id* **80 or 40**.
 - ii. To find the *First_name*, *Last_name*, *Min_Salary*, *Max_Salary*, and *Salary* for each of the employees who are earning more than the maximum salary for that particular job.
 - iii. To show the *First_name*, *Last_name*, *Department_name*, and *City* of employees whose *First_name* contains the letter '**d**'.

Relations	Conditions
Location	All of the attributes should not be null.
Job_Details	The attributes Job_ID and Job_title should not be null and Max_salary and Min_salary must be greater than zero.
Departments	All of the attributes should not be null.
Employee	The attributes First_name, Last_name, and Email must not be null and salary should be greater than zero.

Table 1: Table Constraints

- (f) Write MySQL statements to **prepare the index** for each of the following:
- Create index for the **employee** relation on the attributes *First_name* and *Last_name*.
 - Create index on the attributes *Department_Id* for the **employee** relation.
 - List down the *relational_schema*, *relation_name*, *index_name*, and *column_name* of created indexes for the database **Human_Resource**.
- (g) Write **stored procedures** for each of the following situations:
- To list down the records from each of the relations by passing the **relation name** as the **input parameter**. Run the procedure to display the records for each of the relations in the Human_Resource database.
 - To compute the **total salary** and output it using an output variable.
 - Write a procedure '*Employee_details*' to display the *first name*, *last name*, *email address*, *hire date*, *salary*, and *job title* of the employee for a given employee id. Show the output for the *employee id* **106**.

- (h) **Create triggers** to verify each of the following conditions:
- To assign the *salary* of an employee to the *Max_Salary* of that job type in case where the salary is not ranges between the minimum and maximum salary of that job type while **inserting** a new record to the employee relation. Insert the records in Table 2 to the **employee** relation and display the resulted table.

Table 2: New Employee Records

Attributes	Record 1	Record 2
Employee_Id	126	127
First_name	Rathi	Meena
Last_name	Raman	Mathis
Email	rraman@gmail.com	mmathis@gmail.com
Hire_Date	2003-07-17	2005-10-21
Salary	5000.00	17000.00
Job_Id	AD_ASST	SA_REP
Department_Id	90	80

- To store the **commission amount of 0.5%** of the **current salary** for each employee with **salary more than LKR 10,000.00** before increase the salary. (*Hint: Create new relation Commission (Employee_Id, Commission_amount) to store the commission details for each employee*)
- (i) Write separate transactions for each of the following cases:
- To find the employees with the job id **'IT_PROG'**.
 - To increase the *salary* of each of the **employees** by **0.2%** of the **present salary** and display the records of **Commission relation**.
 - To list down the *First_name*, *Last_name*, *Email*, and *Department_name* for the department managers.

Table 3: Location

Location_Id	City
1000	Colombo
1100	Vavuniya
1200	Anuradhapura
1300	Kandy
1400	Jaffna

Table 4: Department

Department_Id	Department_name	Manager_id	Location_Id
10	Administration	110	1000
20	Marketing	108	1100
30	Purchasing	114	1100
40	Human Resources	103	1000
50	Shipping	121	1000
60	IT	106	1400
70	Public Relations	125	1300
80	Sales	115	1200
90	Executive	100	1300
100	Finance	117	1000
110	Accounting	120	1000

Table 5: Job_Details

Job_Id	Job_Title	Min_Salary	Max_Salary
AC_ACCOUNT	Public Accountant	4200.00	9000.00
AC_MGR	Accounting Manager	8200.00	16000.00
AD_ASST	Administration Assistant	3000.00	6000.00
AD_PRES	President	20000.00	40000.00
AD_VP	Administration Vice President	15000.00	30000.00
FI_ACCOUNT	Accountant	4200.00	9000.00
FI_MGR	Finance Manager	8200.00	16000.00
HR_REP	Human Resources Representative	4000.00	9000.00
IT_PROG	Programmer	4000.00	10000.00
MK_MAN	Marketing Manager	9000.00	15000.00
MK_REP	Marketing Representative	4000.00	9000.00
PR_REP	Public Relations Representative	4500.00	10500.00
PU_CLERK	Purchasing Clerk	2500.00	5500.00
PU_MAN	Purchasing Manager	8000.00	15000.00
SA_MAN	Sales Manager	10000.00	20000.00
SA_REP	Sales Representative	6000.00	12000.00
SH_CLERK	Shipping Clerk	2500.00	5500.00
ST_CLERK	Stock Clerk	2000.00	5000.00
ST_MAN	Stock Manager	5500.00	8500.00

Table 6: Employee

Employee_Id	First_name	Last_name	Email	Hire_Date	Salary	Job_Id	Department_Id
100	Steven	Rajan	srajan@gmail.com	2003-06-17	6000.00	AD_ASST	90
101	Neena	Mathan	nmathan@gmail.com	2005-09-21	17000.00	SA_REP	80

102	Luxi	Kaanthan	lkanthan@gmail.com	2001-01-13	17000.00	MK_REP	20
103	Alex	Ronald	aronald@gmail.com	2006-01-03	9000.00	HR_REP	40
104	Bavan	Kathirvel	bkathirvel@gmail.com	2007-05-21	6000.00	PR_REP	70
105	David	Austin	daustin@gmail.com	2005-06-25	4800.00	FI_ACCOUNT	100
106	Valli	Bala	vbala@gmail.com	2006-02-05	4800.00	IT_PROG	60
107	Diana	Lawrence	dlawrence@gmail.com	2007-02-07	4200.00	AC_ACCOUNT	110
108	Nancy	Desman	ndesman@gmail.com	2002-08-17	12000.00	MK_MAN	20
109	Daniel	Faviet	dfaviet@gmail.com	2002-08-16	9000.00	FI_ACCOUNT	100
110	John	Shekaan	jshekaan@gmail.com	2005-09-28	30000.00	AD_PRES	10
111	Ismael	Mohamed	imohamed@gmail.com	2005-09-30	6000.00	AD_ASST	90
112	Jose	Urman	jurman@gmail.com	2006-03-07	7800.00	HR_REP	40
113	Kamal	Maniyam	kmaniyam@gmail.com	2007-12-07	6900.00	IT_PROG	60
114	Gihan	Perera	gperera@gmail.com	2002-12-07	11000.00	PU_MAN	30
115	Akilan	Thaas	athass@gmail.com	2003-05-18	15000.00	SA_MAN	80
116	Shelli	Baida	sbaida@gmail.com	2005-12-24	3000.00	PU_CLERK	30
117	Swetha	Mohan	smohan@gmail.com	2005-07-24	9000.00	FI_MGR	100
118	Gowri	Balan	gbalan@gmail.com	2006-11-15	6000.00	AD_ASST	10
119	Karan	Kathir	kkathir@gmail.com	2007-08-10	4000.00	MK_REP	20
120	Bandara	Kumara	bkumara@gmail.com	2004-07-18	10000.00	AC_MGR	110
121	Adam	Fripp	afripp@gmail.com	2005-04-10	5000.00	SH_CLERK	50
122	Pathma	Maaran	pmaaran@gmail.com	2003-05-01	7900.00	SA_REP	80
123	Shanth	Velu	svelu@gmail.com	2005-10-10	6500.00	AC_ACCOUNT	110
124	Kevin	Mathu	kmathu@gmail.com	2007-11-16	5000.00	SH_CLERK	50
125	Julia	Nayer	jnayar@gmail.com	2005-07-16	4500.00	PR_REP	70